Group Project Proposal

Counting Reports or Counting Evidence: What Provenance an LLM Orchestrator Needs to Discount Correlated Agent Reports

Group ID: 14

DSA4213: Natural Language Processing for Data Science
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Administrative Information

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Project direction	Whether an orchestrating LLM can aggregate dependent agent reports correctly, and what provenance metadata a multi-agent system must expose for it to do so.
Repository	https://github.com/JasonOng0109/dsa4213-agent-project

Project Summary

Multi-agent systems routinely place one model in the position of aggregating reports from several others. When those agents draw on the same underlying document, their agreement carries no more evidential weight than a single report, but it arrives looking like consensus. We propose a controlled study of that failure and, more importantly, of what repairs it. A reporter model generates lexically distinct paraphrases of synthetic source documents, and an orchestrating judge then answers a factual question from those reports under thirteen conditions that vary report count, message segmentation, agent identity, source independence, and the provenance the judge is shown. In the conflict block, a single fact is added to an opaque baseline, one condition at a time: a named source type, an explicit statement that two reports share a source, a statement of relative source quality, and the last two together. A bare source identifier is tested separately in the agreement block. The primary outcome is the judge's decision against a known world truth. Decision stability across controlled rotations is a secondary measure, and an abstention arm tests directly whether a judge that privately registers uncertainty will act on it. The study aims to decide not whether orchestrators over-count correlated evidence, which prior work already indicates, but **which kind of disclosed metadata actually changes their behaviour**, which is a design constraint for any system that passes provenance between agents.

1. Problem, Research Question, and Motivation

1.1 Research question

When several agents report the same claim, can an orchestrating LLM distinguish the *number of agreeing reports* from the *number and quality of independent documents* underlying them, and what

information must it be given before it does?

The question has two halves, and the design separates them. The first half asks whether the failure occurs at all once the reports are paraphrases rather than duplicates. The second half asks which disclosure repairs it, treating provenance as a graded variable rather than as something that is either present or absent.

1.2 Motivation: why this is an agent problem, not a retrieval problem

In a multi-agent pipeline the orchestrator sees reports, not documents. Agents in one framework commonly share a retriever, a tool, or a base model, so their outputs are correlated by construction, and the architecture turns one piece of evidence into three apparent witnesses before aggregation runs. An orchestrator that reads that as three independent confirmations becomes confidently wrong in exactly the situation a multi-agent design is meant to guard against. A single model reading a single document cannot produce this failure, and it is distinct from an agent misreading its source. Work on governing agent memory states the premise directly: "repetition is not evidence", since agents repeating one claim may represent independent discoveries or "one copied stale note echoed through shared context" [12]. Debate frameworks add pressure from the other direction, since inter-agent sycophancy can collapse a debate into premature consensus before a correct conclusion is reached [11].

The practical question a system builder faces is what to put in the message envelope that passes between agents: a source identifier, a source name, a dependency flag, or a quality score. Each costs something to plumb through a framework, and no published benchmark says which of them changes the outcome. This study answers that question.

1.3 Intended setting

Any orchestrator-plus-workers deployment that aggregates reported evidence: research assistants that fan out sub-queries, triage systems that poll several tools, and debate frameworks that pool agent answers. All expose the aggregator to reports whose dependence structure is invisible in the text.

1.4 Hypotheses

H1 and H2 restate effects that prior work indicates, tested here in a harder setting. H3 to H6 are the study's own targets.

	Hypothesis	Status
H1	Correlated repetition raises the orchestrator's confidence and can beat a single better-sourced report	Expected from prior work, tested here with paraphrases rather than duplicates and against ground truth
H2	The rise in orchestrator's confidence is driven by report redundancy and segmentation rather than by distinct agent identities	Expected from prior work, tested under lexical non-identity
H3	Stating that two reports share a source shifts belief but does not change the decision	Primary target
H4	Stating relative source quality does change the decision	Primary target, with strength of wording as a factor
H5	Judges infer quality from named source types without being told	Secondary, calibrates against prior work
H6	Forced choice conceals a judge's belief state: where its stated probabilities do not identify the answer it gives, either because they sit near indifference or because they favour a different option, an explicit "cannot be determined" option will be taken when one is offered	Primary target

1.5 Definition of success

The project succeeds if, for each of H3, H4 and H6, the condition contrast either shows the predicted effect with consistent direction across judges and cases, or rules it out. A null on H3 is as useful as a confirmation, because it tells system builders that a dependency flag alone does not repair aggregation and that a quality signal has to be carried as well. The project fails only if the instrument is uninterpretable, which means provenance leaking into report text, parse failures, or judges whose stated numbers carry no information and whose decisions are unstable. Sections 3 and 5 specify the checks that detect each of these before analysis.

2. Related Work and Proposed Contribution

2.1 Where this study sits

Our setting is inter-context conflict in the taxonomy of Xu et al. [1]: the conflict is between pieces of supplied evidence, not between evidence and what the model already believes. The neighbouring case, context against the model's own prior, is the subject of ClashEval [2], which perturbs retrieved content and reports that models adopt incorrect retrieved content over their own correct prior knowledge more than sixty percent of the time, with adoption more likely when the model is less confident in its own answer. We deliberately remove that axis by using fictional entities throughout, so that no result of ours can be explained as a prior-versus-context artifact.

2.2 The nearest benchmarks, and what each leaves open

RAMDocs and MADAM-RAG [3] is the closest benchmark in shape. It builds queries whose retrieved documents combine ambiguity, misinformation and noise, and it runs a multi-agent system in which one agent represents each document and an aggregator module combines the debate. It also varies the number of documents supporting each answer from one to three and reports that baselines degrade as that imbalance grows. The evidence unit there is always a distinct document, so report count and source independence move together and cannot be separated. Our design holds the underlying evidence fixed and varies only how it is packaged and attributed, which is exactly the contrast RAMDocs cannot make.

The **Correlation-Aware Memory Arbitration (CAMA) framework** [16] addresses the same failure, which it names *Memory Correlation Bias*: memories written by different agents may inherit the same upstream source, so one evidential factor is counted several times and forms a false majority. CAMA models a retrieved memory slice as query-conditioned latent evidence slots and infers soft memory-to-slot assignments with a set encoder, taking provenance relations as an attention bias rather than a hard dependency label. From the slot presences it estimates an effective number of independent evidence sources through a Hill diversity measure, arbitrates between hypotheses at the level of evidence factors rather than entries, and learns a recovery policy that expands retrieval or traces provenance edges when the independent evidence is insufficient.

CAMA is closer to this project than its memory framing suggests, and the overlap is worth stating plainly. Its correlation-aware benchmark variants are built by paraphrasing and summarising existing memories with provenance links recorded, which is the construction our reporter stage produces, and its Correct Minority Recovery and Replication Sensitivity metrics measure the two effects our X and A blocks measure. The difference is where the correction sits. CAMA gives the system a trained module that infers dependence from the retrieval set and acts on it outside the model, and its comparisons are between aggregation systems. We give the aggregating model nothing but the reports and, in some conditions, one sentence about them, and measure what the model itself does. Whether an orchestrator needs a module of CAMA's kind, or whether a sentence in the message envelope is enough, is the question this design answers and CAMA does not ask.

AuthorityBench [4] measures whether models perceive source authority, operationalised through PageRank over web domains, and finds that authority-guided filtering using the source URL alone improves downstream answer accuracy, while adding the webpage text degrades authority judgements. Two features of that setup leave our question open. Authority is supplied as a property of a source rather than as a statement about the specific reports in front of the model. The benchmark never states or manipulates whether two retrieved documents share a source, and authority is attached to domains rather than to report pairs. It also evaluates open-weight models with the thinking function disabled for the Qwen3 family, which is one reason we specify a reasoning-mode comparison in Section 4.1.

Whose Facts Win? [5] is closest of all, and the paper our design owes the most to. It evaluates thirteen open-weight models on synthetic knowledge conflicts built from fictional entities, using forced-choice prompting, and reports a stable credibility hierarchy in which institutional sources outrank people and social media. It then shows that repetition can reverse that hierarchy. Three of its settings matter for us directly. When two agreeing low-credibility tables are attributed to two different sources, models move toward the repeated claim by an average source-preference gap

of 33.90. When the same content is merged into a single table so that nothing is repeated, models keep their original preference and the gap falls to 6.17. When the two identical tables are attributed to the *same* source, the gap is still 30.04, so the effect survives without any true majority. This is the decomposition of repetition into segmentation and source identity that our A block reproduces, and it is why we treat H1 and H2 as expected rather than as findings.

Minority Sentinel [6] is the nearest multi-agent benchmark to our conflict block because it explicitly studies 2-vs-1 disagreements in which the numerical majority can be wrong and the minority correct. In three-agent debates, the authors show that a numerical majority can suppress a correct minority when agent errors are correlated, and report that roughly one quarter of divergent cases contain a correct minority answer. Their proposed solution is a learned meta-classifier that uses features of the debate trajectory to decide when to overturn majority voting. This motivates our X1 baseline, where two reports support one answer and a single report supports the other: report count alone is not sufficient evidence that the majority is epistemically stronger. The source of correlation is different, however. Minority Sentinel studies correlated agent errors arising from the agents themselves, whereas our design constructs correlation at the *evidence-source* level: two independently generated reports derive from the same underlying source.

FREE-MAD [13] provides a complementary critique of majority-based aggregation. It shows that conformity and error propagation can make consensus unreliable in multi-agent debate, and replaces simple final-round majority voting with anti-conformity prompting and a score-based decision mechanism. Its failure mode is interaction-driven: agents can converge because they influence one another during debate. Our agents need not interact at all. Agreement can instead be misleading because several reports inherit the same underlying evidence. FREE-MAD therefore supports the broader premise that report count should not be treated as evidential weight, while our X block isolates source dependence as a separate mechanism.

Together, these works leave two openings that our design targets. First, the repeated evidence in *Whose Facts Win?* consists of verbatim duplicate tables, and the authors note that deduplicating the knowledge base would solve this simplified case, while realistic deduplication would be neither as trivial nor as appropriate. Our correlated reports are instead independently generated paraphrases of the same underlying evidence, making their shared origin less visible from surface form alone. *Minority Sentinel* likewise identifies correlated majority errors, but does not model whether apparently distinct agents depend on the same external evidence source. Our design therefore makes evidence-source dependence itself the controlled variable.

Second, although prior systems propose alternatives to majority voting and mechanisms for tracking or acting on dependency, they do not directly test whether an LLM aggregator can use *per-instance provenance information* to correct its weighting. *Whose Facts Win?* uses a standing prompt-level instruction to consider source credibility and repetition, while *Minority Sentinel* uses a separately trained meta-classifier based on debate features. In contrast, our X-series keeps the reports fixed and progressively reveals facts about their evidence structure: X3 reveals shared-source dependence, X4 reveals source quality, and X5 provides both. This lets us test not merely whether provenance can be recorded, but whether an LLM orchestrator actually uses it when resolving conflicting reports.

The distinction is where each intervention is applied. *Whose Facts Win?* reports that its prompt-level instruction does not restore the original hierarchy, and that its successful mitigation

is a fine-tuning procedure applied to the model, reducing repetition bias by up to 79.2% while retaining at least 72.5% of original preferences. The four interventions therefore sit in four different places: theirs on the model, the dependency-aware grouping of [12] and CAMA's arbitration on the system, *Minority Sentinel*'s classifier on the debate log, and ours on the message. Ours is neither a retraining nor a system-side aggregation rule, but a per-trial statement about the specific reports shown, supplied at inference. Whether that is sufficient is the open question.

Governing correlated agent traces [12] is the nearest work in premise. It identifies the same causes of spurious agreement, namely shared prompts, shared tools and shared intermediate context, and proposes provenance metadata with dependency-aware grouping as the guard against false promotion of correlated agent evidence, estimating *effective support* by grouping observations according to their dependency structure. It judges raw count and naive source deduplication insufficient, and notes that textual agreement alone is not a calibrated confidence signal. The two studies sit on opposite sides of the model. That work groups evidence by a dependency structure the system already knows, a fix applied outside the model. We measure what a judge does when nobody tells it, and what it does once it is told.

From Agent Traces to Trust [7] provides the provenance-side foundation for X3 and X5. The survey argues that final agent outputs are insufficient for trustworthy aggregation because they do not reveal which evidence supported a claim or what dependencies produced it. It organizes provenance using relations such as SUPPORT, DERIVE, and DEPEND-ON, and identifies multi-agent evidence propagation and semantic provenance through transformations such as paraphrasing and summarization as open challenges. This maps directly onto our reporter stage: two reports may be lexically distinct while still deriving from one evidence item. Our contribution is therefore not to propose another provenance representation, but to test whether an LLM orchestrator can *use* a minimal per-trial provenance statement. X3 reveals only the dependency structure, while X5 additionally states the relative source quality. Their difference lets us distinguish failure to discount correlated report count from failure to know which underlying source deserves greater weight.

2.3 The gap, stated positionally

Figure 1 places the benchmarks that manipulate what the model is shown on the two axes that matter: what form the repeated evidence takes, and what the aggregator is told about it. Each occupies a different corner, and the row this project targets, independently worded reports of one source combined with a per-trial statement about dependence or quality, is unoccupied.

2.4 How prior work shapes each rung of the design

Each contrast in Section 4 exists because some published result makes it necessary to control for an alternative explanation. The tables below map them, the first over the agreement block and the second over the conflict block and the abstention arm.

Contrast	Factor isolated	Prior anchor	What it leaves open
A1 baseline	Belief from one report	Multi-agent debate reports single-model accuracy as its baseline before any peer exposure [8]	No ground-truth-referenced baseline for correlated reporting

Contrast	Factor isolated	Prior anchor	What it leaves open
A1 → A2	Token volume alone	Longer responses win more often in preference evaluation, but the effect tracks information content, not length: copying one response three times leaves information mass unchanged and the win rate *falls*, from 50.0 to 13.0 on AlpacaEval, which the authors attribute to impaired consistency [9]. A pure token-volume account of A1 → A2 is therefore unlikely, though their judgement is a pairwise comparison and ours is not	Whether volume or structure drives the shift when the added text is a paraphrase rather than a copy
A2 → A3	Segmentation into separate messages	Prior exposure to a statement raises its later truthfulness rating in an LLM [10], and merging repeated tables into one removes most of the effect [5]	Whether segmentation still matters when the segments are not identical strings
A3 → A4	Distinct agent identities	Debate protocols treat agreeing outputs as corroboration [8], work on consensus-free debate argues that an LLM judge aggregating agent outputs is prone to conformity and so effectively equivalent to majority voting [13], and repetition from one source already flips preferences [5]	Whether identity adds anything once segmentation is held fixed
A4 → A5	Bare source identifier	Provenance metadata with dependency-aware grouping is proposed as the guard against false promotion of correlated agent evidence, with raw count and naive source deduplication explicitly judged insufficient [12]	Whether an identifier is read as structure or ignored
A5 → A6	Explicit source dependence declaration	CAMA identifies Memory Correlation Bias [16], where shared upstream sources create a false majority, and combines neural dependency inference with provenance priors to estimate the effective number of independent sources, establishing that reliable aggregation must model dependence rather than count entries	Whether a LLM judge can natively perform this factor-level arbitration purely through explicit, in-context metadata
A4 vs A7	True independence	RAMDocs varies supporting document counts, but every document is distinct [3]	Correlated and independent agreement at equal report count

Contrast	Factor isolated	Prior anchor	What it leaves open
X1 → X2	Named source type	Models perceive authority from source identity alone, and filtering on it helps [4], with a stable type hierarchy [5]; Minority Sentinel shows that a 2-vs-1 majority can suppress a correct minority [6]	Whether that perception survives a 2-vs-1 report count

Contrast	Factor isolated	Prior anchor	What it leaves open
X3 vs X4 vs X5	Dependence against quality	Standing credibility instructions are insufficient under repetition [5]; provenance work calls for tracking cross-agent dependency and evidence propagation [7]	Which per-trial statement moves the decision
Abstention arm	Acting on uncertainty	With a "cannot be answered" option, most models still answer [5]	Whether a judge at indifference abstains once its stated belief is also elicited

2.5 Proposed contribution

The single claim the design is built to decide is that **disclosure about sources changes what an aggregator believes, while only a statement of relative quality changes what it chooses**. The contributions below are ordered by how directly they serve that claim.

1. **The disclosure dissociation.** No prior benchmark states the dependence structure to the model per trial and measures the effect, nor separates that from stating quality. Conditions X3 (dependence), X4 (quality) and X5 (both) isolate them, and the contrast between them is the project's main result whichever way it falls.

2. **A belief-versus-choice measurement.** The nearest prior work reads token probabilities under forced choice, so it cannot observe a judge whose stated probabilities do not identify the answer it gives, whether because they sit near indifference or because they favour a different option. We elicit a decision and a probability for every option, and add an abstention arm, so the gap H6 names is measured rather than inferred.

3. **A correlated-evidence instrument that deduplication does not remove.** Our correlated reports are independently generated paraphrases that pass a semantic-equivalence check and a lexical-overlap filter, with provenance confined to a structured field the reporter never sees and a static linter enforcing the authoring rules. *Whose Facts Win?* names this case as untested, and CAMA builds paraphrase-derived correlated memories but supplies them to a system-side method rather than showing them to a judge under controlled disclosure.

4. **Frontier orchestrators with a reasoning-mode comparison.** The nearest benchmarks use open-weight models, and the authority work disables the thinking function for its Qwen3 models, so this comparison is unoccupied. Section 4.1 describes it as planned rather than promised, since it is one of the ranked optional sub-experiments in Section 6.

5. **A quantity, not only accuracy.** From a judge's probabilities on a no-evidence condition, a single-report condition and a three-correlated-report condition we estimate the number of independent sources the judge behaved as if it saw, and compare that against the genuinely independent condition. Section 4.2 gives the estimator and Section 5 lists the conditions it is computed from. Related work estimates effective support by grouping evidence according to a dependency structure the system already knows [12], and CAMA estimates a quantity of the same name from a learned assignment of retrieved memories to latent evidence slots [16]. Ours

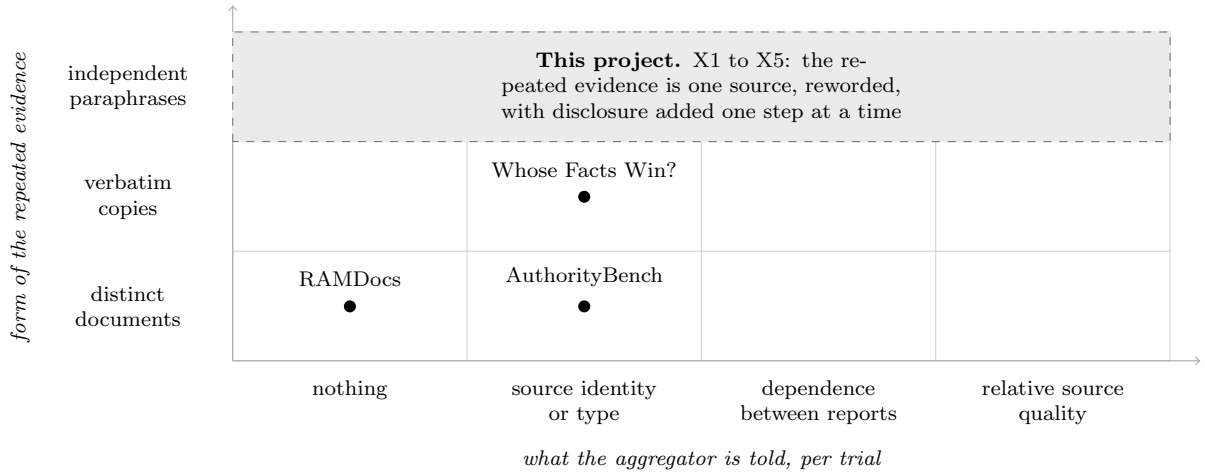


Figure 1: Where the nearest work sits. Minority Sentinel, CAMA and FREE-MAD are not plotted because each intervenes outside the model, through a meta-classifier, a learned arbitration module and a revised voting rule, rather than by varying what the aggregator is told. ClashEval is not plotted because it varies a third axis, the model's own prior against retrieved context, which this design removes by using fictional entities. The shaded region is the target: correlated evidence that deduplication cannot remove, combined with disclosure that is stated per trial rather than as a standing instruction.

is a behavioural measure of the model itself: what a judge does when nobody tells it, rather than a correction applied outside it.

2.6 What we will not claim

That repeated lower-quality sources beating a single higher-quality one is new, that decomposing repetition into segmentation and identity is new, or that majority-following in LLMs is new. Each has a prior anchor cited above, which the write-up states next to the corresponding result. Our contribution is the disclosure dissociation and the belief-versus-choice measurement, not the bias itself.

3. Agent and Task Setup

Task. A question with a known answer and a fixed set of answer options, presented with one to three reports from named agents. The orchestrator returns a chosen answer and a probability for every option. The corpus uses four options per case: the world truth, the claim made by the weaker source, and two plausible values that no source supports. Every case therefore asks an open-valued question, since a yes/no question cannot carry distractors. The option set is a case-level property stored in the trial record rather than a property of any condition, and the trial builder, not the case, decides where each option sits.

Environment. A closed synthetic corpus with no live retrieval. Each case fixes a world truth, one weaker source whose claim is wrong, and three independent higher-quality sources that support the truth. All entities are fictional, so no orchestrator can answer from prior knowledge, and cases are authored across three domains (institutional logistics, product specifications, and incident or status reports) so that domain can be checked as a nuisance factor rather than assumed away.

Source quality is carried by one of four dimensions, which are authority, directness, method, and primary-source status, recorded per case so that results can be broken down by dimension. Type names are authored so that the name itself carries the case's quality dimension: the authoring test is that a reader shown only the two type names, with all content stripped, would prefer one of them. Where a dimension cannot be carried by a name, the case is marked as testing the other hypotheses and H5 is reported only over the cases where the name does carry it. Three domains and four dimensions do not divide forty evenly, so the design is unbalanced by construction rather than fully crossed, and domain and dimension are analysed as unbalanced factors rather than read as a balanced grid.

The source repeated in the agreement block is always the strong one, and the weak source appears only in the conflict block. The independence index of Section 4.2 is read on the probability of the world truth, so the repeated evidence must support the truth: with a weak shared source, the one-report and three-report conditions would push belief in the truth down while the three-independent-sources condition pushed it up, and the index would invert sign against its normative value. Fixing the shared source removes that problem and one authoring decision per case. The agreement block therefore measures belief, and the conflict block, where counting reports gives the wrong answer, measures decisions. The weak source needs two accepted paraphrases, the strong shared source three, and the other two strong sources one each.

Two agent roles. A *reporter agent* converts one source document into report text. An *orchestrating judge* reads the reports and decides. The judge receives the reports, the answer options, and whatever provenance the condition specifies, is not asked for reasoning in the main arm, though some models may reason regardless (Section 6), and returns a machine-parseable JSON object.

The reporter stage runs once per source and is the step that makes correlated reports usable as an instrument, so its procedure is fixed in advance:

1. **Generate.** Provenance is a structured field of the case file that the reporter never receives: it is given the source's content only, and writes a short faithful report. This structural separation is the primary defence against provenance leaking into report text, the filter in step 2 is a backstop against the reporter inventing an attribution, and the static linter checks the case file. Sampling runs at high temperature, because three reports of one source must be worded differently and low-temperature sampling returns near-identical sentences.

2. **Check each candidate** against three acceptance criteria:

   (a) semantic equivalence to the source, judged by an independent rater;

   (b) pairwise lexical overlap with the other selected candidates below threshold;

   (c) no word that appears only in the source's type description, and no invented attribution phrase.

3. **Regenerate on failure**, never hand-edit, so the frozen text stays attributable to the reporter model's own output. Each round generates six candidates per source, and there are at most two rounds; the second re-rates the first round's candidates alongside its own and selects over the pooled set. A source still short after two rounds stops the run rather than lowering a threshold.

The sampling temperature therefore governs how many candidates are discarded, not which are admitted, and the acceptance rate is reported.

4. **Freeze.** Accepted reports are written to disk and reused verbatim, so every judge sees identical text and reporter stochasticity cannot contaminate a comparison between judges.

The confidence screen of Section 5 is the one exception to that pipeline. Its three statements are authored case fields rather than reporter output, because they are probes of a judge's stated numbers rather than reports from a source, so they sit outside the acceptance filter and the never-hand-edit rule that governs report text.

Models. All access is through OpenRouter, and every trial record stores the model identifier, since frontier model behaviour is a moving target. Each judge is pinned to a single serving provider, and the provider is recorded in every trial record, so that a difference between conditions cannot come from the router sending them to different hosts of the same model.

Role	Model	Why this one
Reporter	A smaller instruction-tuned model	Cheap, and drawn from a family not used as a judge, so no judge grades its own writing style
Primary judges	Claude Sonnet 5, GPT-5.4, Claude Opus 5	Span two developers and a range of calibration behaviour, so a pattern present in one model is reported as model dependence
Auxiliary models	Qwen 3.6 27B and Llama 4 Maverick, open-weight models that return a full token distribution	Never judge a main-arm trial. Used only to validate the stated-confidence measure against the model's own probabilities

Inputs and outputs. The input is a rendered prompt containing the question, numbered options, the reports, and an optional provenance note. The output is a JSON object with the chosen answer as a verbatim string and a probability per option. Options are numbered rather than lettered, following the nearest prior work, so that answer labels cannot be confounded with the indices of the evidence items [5].

Protocol and constraints. Judging runs at temperature 0. Presentation order, the slot holding the independent report, the position of the world truth among the options, and the offset of the wrong claim from the truth are rotated on a fixed schedule rather than shuffled, so each position is covered a known number of times. The wrong-claim offset rotates separately from the truth's position because rotating the option list as a block would keep the truth and the wrong claim adjacent in every trial. Option order therefore belongs to the trial builder, not the case: an author supplies the options and the rotation sets where the truth and the wrong claim sit. That leaves the static linter only hard assertions to check. Four concern the answer space: that the world truth is among the options, that exactly one source supports a wrong answer, that exactly two distractors are supplied, and that any option no source supports is claimed by no report. Two are leak assertions on the case file itself: that no word appearing only in a source's type description

appears in any source's content, and that no supersession, staleness or hearsay phrasing appears in the content of the competing pair. These are the third of the three defences against provenance leakage, alongside the structural separation and the acceptance filter.

Stopping conditions and failure recovery. Total API spend is capped. A pre-flight call estimate is printed before any run, and every run has stated stop conditions so that a misconfigured job cannot drain the budget. Judge outputs that do not parse are logged with an explicit parse-failure flag and excluded from decision denominators rather than dropped silently or guessed. A leak check over the frozen reports runs before any judging, and a failed check blocks the run.

4. Proposed System or Study Design

Pipeline. Figure 2 gives the order of operations. Everything downstream of the source document is frozen once, so the only thing that differs between conditions is what the judge is shown.

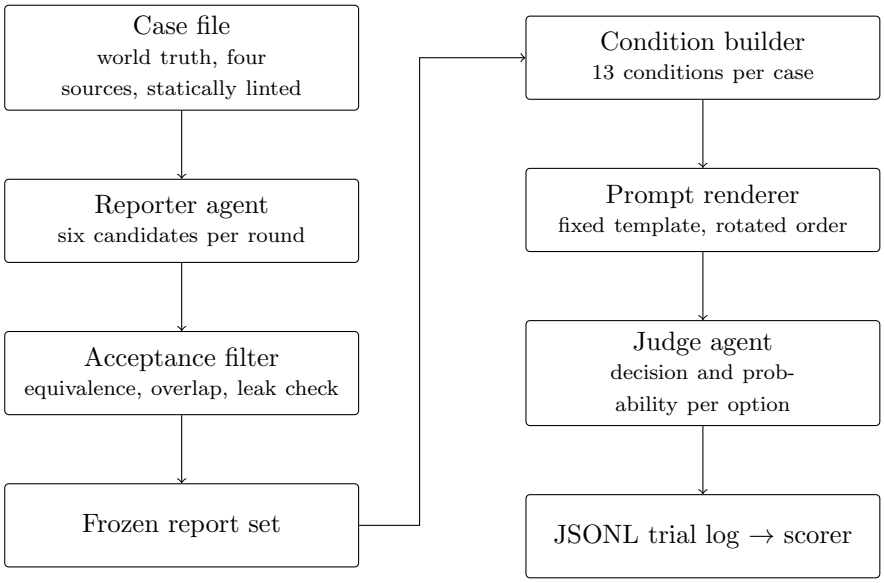


Figure 2: The three-stage pipeline. Reports are generated and frozen once per case, then reused verbatim across every condition, judge and repetition, so reporter stochasticity cannot contaminate a between-condition comparison. Candidates that fail the acceptance filter are regenerated rather than edited.

The condition matrix. Thirteen conditions in two blocks. Each condition is defined by a factor vector, which is stored in the trial record so that analysis groups by factor rather than by condition name.

ID	reports	msgs	agents	sources	conflict	provenance shown	depend. stated	quality stated
A0	0	0	0	0	no	none	no	no
A1	1	1	1	1	no	none	no	no
A2	3	1	1	1	no	none	no	no
A3	3	3	1	1	no	none	no	no
A4	3	3	3	1	no	none	no	no
A5	3	3	3	1	no	opaque ID	no	no
A6	3	3	3	1	no	opaque ID	yes	no
A7	3	3	3	3	no	none	no	no
X1	3	3	3	2	2-vs-1	none	no	no
X2	3	3	3	2	2-vs-1	named type	no	no
X3	3	3	3	2	2-vs-1	none	yes	no
X4	3	3	3	2	2-vs-1	none	no	yes
X5	3	3	3	2	2-vs-1	none	yes	yes

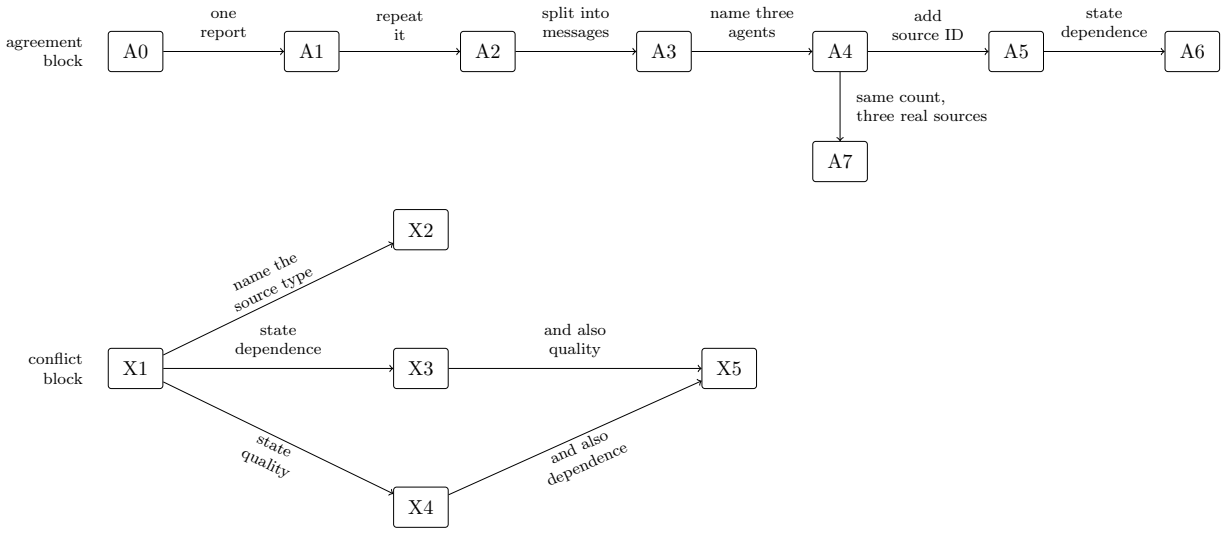


Figure 3: The condition ladder. In the agreement block each arrow changes exactly one factor, building up from no evidence to three correlated reports with the dependence disclosed. The conflict block is a set of single additions to the X1 baseline, with X5 combining two of them, so it is drawn as a hub rather than as a chain. Throughout it, the same two-against-one evidence is held fixed and only what the judge is told about it varies, with X3 against X4 separating a statement of dependence from a statement of quality.

This matrix is the planned core of the study. Conditions may be added if an early configuration run exposes an alternative explanation it does not control for, and any addition is reported as such. Figure 3 names the factor each step moves. Two conditions carry extra weight. A0 shows the question and options with no reports, which establishes the judge's prior, gives the estimator in Section 4.2 its origin, and doubles as a sanity check, because a judge with no evidence should sit near chance, which with four options is 25%. The two blocks disclose dependence differently, and deliberately so. A6 states it on top of the identifier A5 introduces, because the agreement ladder adds one element at a time. X3 states it with no identifier shown, so that X1 against X3 measures the statement itself rather than the statement plus an identifier. The statement names agents rather than sources, so it stands without one. X1 is the opaque conflict baseline, in which two paraphrases of the weaker source support a wrong answer against one report from the stronger

independent source, and every other conflict condition adds exactly one kind of information to it, except X5, which combines two so that their joint effect can be compared with each alone.

4.1 Comparisons and Ablations

- **The condition matrix itself** is the primary controlled comparison, with one factor moving per step, and every comparison is reported per judge before any pooling.
- **Report length**, as a trial-level factor. Each report is either the accepted paraphrase alone or the same paraphrase embedded in authored, non-decisive filler, giving long reports of 88 to 96 words, with the longest report in a trial at most 1.10 times the shortest, so that length cannot substitute for provenance as a reason to prefer a report. The band and the cap are mutually consistent, since 96 divided by 88 is 1.091. Because the filler is authored and the paraphrase is shared between both variants, any difference is attributable to length, which published work identifies as a confound in evaluation settings generally [9]. The factor crosses A1 and A2 as well as the conflict block, because H2 asks whether the rise from A1 to A2 tracks token volume or redundancy: a larger rise at long length would point to volume, an equal rise to redundancy as such.
- **Abstention arm.** The conflict block, plus the three-independent-sources condition A7 as a per-judge baseline, is repeated with an explicit "cannot be determined from the evidence provided" outcome. Prior work reports that most models decline such an option even when the evidence does not settle the question [5]. Models also differ hugely in how readily they take it, so an abstention rate only means something next to how often the same judge abstains when the evidence is clear, which A7 supplies. We pair the arm with an elicited probability per option, so a judge whose stated probabilities do not identify the answer it gives – whether they sit near indifference or favour a different option – is visible in a single record.
- **Quality-statement strength**, a graded ladder that runs in the main programme rather than as optional work, because it is H4's only defence against its most obvious alternative explanation. The X5 wording is categorical, which risks measuring compliance rather than reasoning. Three rungs vary only an intensity phrase, from somewhat to considerably to far more reliable than the shared source, turning the ladder into a dose-response curve with the categorical wording as its ceiling. If the weaker wordings also move decisions the judge is weighing evidence; if only the strongest does, it is complying with an instruction that states which source to prefer. It runs on 10 cases, the same ten as the length arm so the two subset arms can be cross-referenced, with twelve repeats and all three judges, since H4 is a claim about decisions and a judge whose stated confidence is unusable still supplies usable decisions.
- **Reasoning on and off**, crossing reasoning mode with report length on X1, X3 and X5 for two cases, one model in both modes. Toggling the provider's reasoning parameter keeps the comparison within a model rather than across models. Traces are logged and read for whether the model mentions the shared source before deciding, which separates not noticing the dependence from noticing it and not acting on it. Which model can serve as the "off" arm depends on whether it can be held to the no-reasoning protocol (Section 6).

- **Confidence-measure validation.** Verbalized probabilities are compared against option-index token probabilities on the two auxiliary models, which never judge a main-arm trial. None of the primary judges returns a distribution over the option tokens, so this is a validation study of the measure rather than a check on any judge. Auxiliary models are selected for exposing token probabilities, not for the quality of their stated confidence, because the check asks whether written numbers track internal ones – a question best posed where the two may diverge. Because it runs on models that are not our judges, it speaks to written confidence as a class of measurement, not to any particular judge's numbers.

All conditions state provenance per trial, about the specific reports shown. This is distinct from a standing instruction to weigh credibility or discount repetition, which prior work uses as a baseline method and reports to be insufficient [5].

4.2 Quantifying the effect: effective independent sources

Accuracy per condition says whether an orchestrator over-counts correlated reports, but not by how much. From the judge's stated probabilities we estimate the number of independent sources it behaved as if it saw. Under the standard account, independent evidence adds in log-odds, so with $l(c)$ the logit of the judge's stated probability on the world truth in condition c:

$$n_{\text{eff}}(c) \;=\; \frac{l(c) - l(A0)}{l(A1) - l(A0)}$$

Normatively n_{eff} is 1 for A4, which shows three reports from one source, and 3 for A7, which shows three reports from three sources. So $n_{\text{eff}}(A4) - 1$ measures credited independence that does not exist, and $n_{\text{eff}}(A4) - n_{\text{eff}}(A6)$ measures how much a dependence statement claws back. This gives a continuous form of H3 that a judge can move without crossing a decision boundary. The estimator reads the probability of the world truth, so it requires the repeated evidence to support the truth; this is why the agreement block always repeats the strong source (Section 3), and repeating the weak one would make the index come out negative.

The estimator requires the denominator to be bounded away from zero. Where one report moves a judge by less than 0.2 logits – a shift from 25% to 28.9% – the index is reported as undefined rather than as a large number, and the proportion of undefined cells is reported per judge. The confidence screen previews this: its uninformative and determinate levels approximate the no-report and one-report conditions, so a judge with a narrow gap there will not yield a usable index.

This is not a measure we invent. Correlation neglect, the failure to account for possible repetition in the information one receives, is formalised by DeMarzo, Vayanos and Zwiebel [14] and shown experimentally by Enke and Zimmermann [15], in both cases for human decision makers. Applying it to an LLM orchestrator as a measure of the model's own behaviour, on a benchmark built so the quantity is identifiable, is the contribution. It differs from estimating effective support over a dependency structure the system already knows [12], and from CAMA's effective number of independent sources, a Hill diversity measure over learned slot assignments in a retrieval set [16]. Ours is read off a judge's own stated probabilities, so it indexes the behaviour of the aggregating model rather than the structure of the evidence it was given. It is reported only for judges that

pass the screening in Section 5, and indexes behaviour on each judge's own stated scale rather than internal probabilities.

5. Preliminary Evaluation Plan

Claim-to-evidence mapping.

Claim	Evidence
H1 amplification	A1 against A4 for the confidence rise, read on the decision and the probability of the world truth per judge and case; and X1's decision rate on the repeated claim, against the 25% chance floor, for whether repetition beats the better-sourced single report. The second half is a level rather than a contrast between conditions
H2 decomposition	A1 to A2 to A3 to A4, one factor per step, and A4 against A7 for true independence
H3 dependence insufficient	X1 against X3 decisions, with X3 against X4 separating dependence from quality
H4 quality sufficient	X1 against X4 against X5, with wording strength as a graded factor
H5 inference from names	X1 against X2, over the cases whose type names carry the quality dimension (Section 3)
H6 belief and choice gap	Observation, in the forced-choice arm: the rate of records whose stated probabilities do not identify the answer given, either because the answer is not the highest-probability option or because the margin between the top two options falls below a stated threshold. Test, in the abstention arm: the abstention rate in those same trials, within judge and read against that judge's own abstention rate in A7. Decision stability supports the reading rather than defining the hypothesis
Independence index (Section 4.2)	n_{eff} from A0, A1, A4, A6 and A7 probabilities, reported only for judges that pass the screening step below

X1 appears in three rows without being double counted: it is the conflict block's opaque baseline, so H1 reads its accuracy as a level, H3 reads X1 against X3, and H5 reads X1 against X2.

While H3 is conceptually framed by the **A5** $\rightarrow$ **A6** contrast, we test it using **X1** $\rightarrow$ **X3**, under conflict. In the conflict-free A5 and A6 the judge faces no competing answer, so a dependence statement has no decision to change and the contrast cannot separate H3 from indifference. X1 and X3 supply an active 2-vs-1 conflict, which is the setting in which explicit dependency metadata either changes the decision or fails to.

Precommitted interpretation. The table below fixes, before any data is collected, what pattern counts as support, what counts as a clean null, and what would indicate that a nuisance

variable rather than the manipulation is driving the result. It is agreed by all members before collection begins, is not revised after seeing data, and any departure from it is reported as such.

Comparison	**Counts as support**	**Counts as a clean null**	**Indicates a nuisance variable**
A1 → A4 (H1)	Belief in the repeated claim rises in the same direction in most case-by-judge cells, for every judge that passes confidence screening	No consistent direction across judges and cases	The rise appears only in one domain, or only when the repeated reports occupy particular positions
A2, A3, A4 (H2)	A3 and A4 exceed A2 and differ little from each other	All three within noise of one another	The A1 to A2 effect is larger at long report length than at short, which the length arm detects because it covers A1 and A2 as well as the conflict block
X1 → X3 (H3)	Belief moves toward the truth while the decision rate is unchanged within repetition noise	Neither belief nor decision moves	Movement occurs only in cases where a mild content cue survives, and not in the near-symmetric cases
X3, X4, X5 (H4)	X4 and X5 change decisions where X3 does not	X4 and X5 behave like X3	Decisions change only under the strongest wording, which the graded ladder identifies as compliance rather than reasoning
X1 → X2 (H5)	Accuracy rises with a named source type alone, with no quality statement	No change from X1	The gain appears only for source types the judge could match to a familiar real-world institution
Abstention arm (H6)	Abstention concentrates in the trials whose forced-choice probabilities do not identify the answer given, whether by a margin between the top two options below the stated threshold or by the answer differing from the highest-probability option	Abstention rate is flat across conditions, including against the A7 baseline	One judge abstains everywhere or nowhere, which that judge's own abstention rate in A7, where three independent sources agree, detects

Measures and how much weight each carries. Conclusions rest on what the judge did, and its stated numbers only support them. The order below is fixed in advance because the measures can disagree, and H3 predicts exactly that case.

	Measure	How it is used
1	Decision against world truth	Primary. Every claim in the write-up rests on it
2	Decision stability across rotations	The proportion of repeats in a judge × case × condition cell that return the most common answer: 1.0 where every repeat gives the same answer. The floor is not the 0.25 chance of any single option, because the measure takes the largest of four counts: under random answering the expected value is about 0.41 at twelve repeats, 0.47 at six and 0.53 at four, so the baseline is simulated per repeat count rather than assumed. Reported with the repeat count, since stability over four repeats and over twelve are not comparable. Requires no self-report, so it survives a judge whose stated numbers are disqualified
3	Stated probability	Analysed within judge as differences and ranks, never as a calibrated quantity across judges
4	Abstention rate	Analysed within judge against that judge's own no-conflict baseline in A7, because propensity to take a neutral option varies widely between models [5]

Screening stated confidence. Stated confidence is screened rather than trusted, because a judge that returns a high number on every answer would make the measure uninformative without that being visible in the numbers themselves. A stated probability is admitted as evidence only if it passes both checks below, each run and reported per judge whatever the outcome.

Check	What it requires	How it is assessed
Ordering	The numbers rise across three single-report trials of increasing informativeness: an irrelevant report, one narrowing the options to two, one naming the answer	Rank correlation and a test for ordered alternatives, not inspection
Discrimination	The stated probability separates correct from incorrect decisions	Area under the ROC curve, where a value near 0.5 means the number carries no information, computed within condition, since pooling across conditions would conflate a judge's confidence with the manipulation itself

A judge that fails is reported as uninformative on confidence and excluded from probability-based analysis, with its decisions still counted. Whether a written number tracks the model's own token probabilities cannot be checked on any primary judge, since none returns a distribution over the option tokens; it is run instead as the confidence cross-check among the optional sub-experiments (Section 4.1), a validation of the measure rather than a screen. Calibration summaries

such as the Brier score are reported only where stated values have usable spread, since a judge sitting at 95 to 100 throughout makes them degenerate.

Handling variability. Rotation covers presentation order, independent-report slot, the world truth's option position, the wrong-claim offset, and cue position in the long-report arm, with repeats chosen so each rotation is covered a stated number of times and any residual imbalance reported. Twelve repeats give exact balance over (slot, truth position) and (truth position, wrong offset), and complete but uneven coverage of (slot, wrong offset), where three of the nine combinations appear twice; full crossing of all three factors would need thirty-six repeats, so we claim pairwise balance, not full crossing. The other two schedules are unbalanced on purpose and the argument for each is stated rather than assumed. The agreement block's six repeats give two of the four truth positions double weight, but the schedule is identical across every agreement condition and every claim in that block is a contrast between conditions, so a constant position bias cancels. The abstention arm's four repeats cover all four truth positions exactly once, which is the factor that matters for a rate computed over options; its coverage of the independent-report slot is uneven, and because an abstention rate is a level rather than a contrast that imbalance does not cancel, so the slot coverage is reported beside the rate. The main run uses one generated report set. A wording-robustness check that regenerates the reports for two cases and re-judges the conflict block is one of the ranked optional sub-experiments in Section 6, and if it is not run the limitation is stated rather than papered over.

Run plan and cost. The study runs on 40 cases, each passing through up to four arms: the *short arm*, all 13 conditions with the accepted paraphrases alone; the *length arm*, the conflict block plus A1 and A2 with long reports; the *abstention arm*, the conflict block plus A7; and the *confidence screen*, three single-report trials per answer position. The three primary judges run every trial of the short arm, the abstention arm and the screen. The length arm is a robustness check rather than a primary claim, so it runs on 10 of the 40 cases with two of the three primary judges, Claude Sonnet 5 and GPT-5.4: one per developer, so the arm keeps the cross-developer comparison the study relies on, and a pair that differs in both ways length interacts with, since in our sample case one gave graded confidence and reasoned on conflict trials while the other sat at the top of its confidence scale and did not reason. It covers the conflict block, where H3 and H4 live, and A1 and A2, where H2 lives, so that the length factor crosses the contrast it is named as the control for. The graded wording ladder of Section 4.1 runs on the same 10 cases, with three rungs, twelve repeats and all three judges. The two auxiliary models judge no main-arm trial and appear only in the optional confidence cross-check. The agreement block uses six repeats, because its comparisons are movements in belief, and the length arm's A1 and A2 cells use six to match it; the conflict block uses twelve, so that every pairing of the truth's position (four values) with the wrong-claim offset (three) occurs once; the abstention arm and the screen use four.

Costs are derived from a single sample case authored and generated end to end, multiplied to the target. Calls are priced at three rates, because they are not alike: report generation and rating at \$0.00025 per call, measured; judging short reports at \$0.00178 per call, measured and blended over the three primary judges (Claude Opus 5 \$0.00293, Claude Sonnet 5 \$0.00128, GPT-5.4 \$0.00114); and judging long reports at \$0.0030 per call for the Sonnet and GPT-5.4 pair, the one figure estimated rather than measured, conservatively set at roughly 2.5 times that pair's measured short-report rate.

Per case, arms run on all 40 cases	**Calls**	**Cost**
Generate and accept paraphrases: 4 sources × 12 calls	48	$0.01
Short arm, agreement block: 8 conditions × 6 repeats × 3 judges	144	$0.26
Short arm, conflict block: 5 conditions × 12 repeats × 3 judges	180	$0.32
Abstention arm: 6 cells × 4 repeats × 3 judges	72	$0.13
Confidence screen: 3 levels × 4 rotations × 3 judges	36	$0.06
Per case	**480**	**$0.78**

Programme	**Cases**	**Judges**	**Calls**	**Cost**
All arms above	40	3	19,200	$31.24
Length arm, conflict block: 5 conditions × 12 repeats	10	2	1,200	$3.60†
Length arm, A1 and A2: 2 conditions × 6 repeats	10	2	240	$0.72†
Graded wording ladder: 3 rungs × 12 repeats	10	3	1,080	$1.92
Main programme			**21,720**	**$37.48**
Optional sub-experiments (Section 6)	subsets	varies	1,112	$2.78
Total			**22,832**	**$40.26**

† At the estimated long-report rate; every other figure uses a measured rate.

If every source needs a second generation round, the worst case adds 2,880 generation calls and $0.72. The programme costs roughly $40. We request $75, which covers the long-report rate being estimated rather than measured, and one full rerun of the main programme. If the long-report rate proves higher than estimated, the length arm is the first thing trimmed.

Representative evidence. The write-up will include one fully rendered prompt and judge response per block. Where trials in which the stated probability and the chosen answer disagree occur, one is shown in full, since that single record displays the strong form of the state H6 names, in which the stated belief favours an answer other than the one given. If none occurs, their absence is reported as evidence against H6 rather than passed over. Cases whose paraphrases fail the acceptance thresholds are excluded from interpretation rather than pooled, and the exclusions are listed.

6. Risks, Limitations, and Responsible Practice

Design risks.

- *Provenance leaking into report text* would invalidate every opaque condition. The three defences in Section 3 apply, and a leak check over the frozen reports blocks the run if it fails.
- *Judge ceiling effects* would make confidence measures uninformative. This is handled by the screening step in Section 5 and by treating the decision as primary.
- *Position and option-order confounds* are controlled by rotation rather than shuffling. Adding a second position factor in the long-report arm requires more repeats, and the residual imbalance is reported.

- *The answer space has four options, not two.* Four is not a convention: the nearest work uses two options, yes/no or free text, and AuthorityBench chose yes/no questions deliberately, to enable accurate evaluation of answer correctness [4]. We choose four for two reasons. At two options a judge that follows the report count and a judge that ignores the reports produce the same answer, so the two cannot be told apart; with two options unclaimed by any source, the chance floor falls to one quarter and careless selection becomes visible. And the confidence screen in Section 5 needs a middle rung, a report that rules out two options and leaves two, which two options cannot provide. The cost is that naturally binary questions are excluded, so every case must ask a question with more than two sensible answers. Option order is rotated so that a fixed answer position cannot substitute for reading the reports.
- *A judge may reason when it is not asked to.* Frontier judges may reason adaptively even when reasoning is not requested, and may do so unevenly across conditions, which would make a judge a different instrument in one block than in another. Reasoning tokens are logged per trial, suppression is attempted explicitly, and any judge that cannot be held to the no-reasoning protocol is reported as such rather than presented as comparable. The response budget is set high enough that reasoning cannot truncate an answer.

Scope limitations.

- Each reporting agent is given exactly one source. Real pipelines usually have each agent read several sources, so their reports overlap partially rather than completely. We fix one source per agent because it is the only way to move report count and source independence independently, and partial overlap is at least as hard a case rather than a different one. The nearest published benchmark makes the same simplification, assigning one agent per retrieved document [3].
- Fictional entities and synthetic sources buy control at the cost of ecological validity. The study says what orchestrators do with a clean manipulation, not what they do on live retrieval. This is the same trade the nearest prior work makes and defends [5].
- Frontier model behaviour is a moving target. Model identifiers are recorded per record and results are tied to dated versions.
- A quality-disclosure condition stated in strong wording risks measuring compliance rather than reasoning. This is addressed rather than merely noted, through the graded wording ladder of Section 4.1, which runs in the main programme rather than as optional work, with the categorical wording reported as a ceiling reference rather than as the result.
- Four optional sub-experiments are scoped, costed and ranked: the reasoning comparison first, at 144 calls, because it is the one unoccupied comparison in this literature and its traces show whether a judge notices the shared source at all; then a wording-robustness rerun at 392 calls, the confidence cross-check at 360, and the independence index at long report length at 216, together 1,112 calls at about $2.78. That total is priced at an assumed $0.0025 per call, above the measured short-report rate because the reasoning comparison bills reasoning tokens as output. The ranking reflects what the plan learns, not only what it costs. One of the four is conditional on results rather than on money: the independence index at long length requires the index to be computable at short length, which requires at least one judge to pass the confidence screen. The

long-report rate is also estimated rather than measured, so the total may move once the length arm runs. They run in priority order, and the write-up states which were executed and at what scale.

- Two of the three primary judges come from one developer, and error correlation between related models is documented, which limits claims about generality.
- This study places all of its weight on the judge, and measures one aggregation step rather than a full pipeline. Our question is what information changes the judge's own aggregation, not whether a judge-side fix changes end-to-end system accuracy. If fixing the judge turns out not to fix the system, that is a finding about where the fix belongs, and one this design cannot deliver.

Novelty risk. The closest prior work already covers repetition beating credibility and the segmentation and identity decomposition [5]. Work on correlated agent traces already states that repetition is not evidence and estimates effective support by grouping evidence according to a known dependency structure [12]; our independence index measures instead what a judge does when that structure is not given. Our contribution has to be stated as the disclosure dissociation and the belief-and-choice measurement, not as the existence of the bias, and Section 2.6 fixes that wording in advance.

Responsible practice. There are no human subjects, no personal data and no private documents, because every source is written for the study, and the fictional institutions, products and incidents are not usable as misinformation about real entities. API keys live in a local environment file and never in the repository, and spend is capped, estimated before each run and approved explicitly. We will release the case files, the frozen prompt templates, the generated report sets with their acceptance scores, and the full trial log, so the study is reproducible from the released artifacts alone.

Appendix A. A worked trial

The example below is condition X3 on the sample case, at one point in the rotation schedule: the lone strong report is read third and the world truth sits last in the option list. All four options rotate jointly, so the list is not sorted and its order carries no information about which value was moved; sorting the options would make the truth's position deterministic and defeat the rotation. Two reports derive from the weaker source and support the wrong answer, one report comes from the stronger independent source and supports the truth, two options are distractors that no report supports, and the judge is told which two reports share a source but not which source is better. The report strings are frozen output from the reporter stage. The response block shows the required output format; its values are illustrative, since collection has not run.

```
--- prompt as rendered ------------------------------------------------
You are asked to answer a factual question based on reports from others.
Read all reports below, then give your answer.

Question: What percentage of the final ME2103 grade comes from the lab logbook?
Possible answers:
  1. 10%
  2. 25%
  3. 30%
  4. 15%

Agent A says: A 25% portion of the final grade for module ME2103 is determined
              by the lab logbook.
Agent B says: The final module grade for ME2103 is comprised of 25% from the
              lab logbook.
Agent C says: The lab logbook accounts for 15% of the final grade for the
              ME2103 module.

Note: Agent A and Agent B obtained their information from the same underlying
source; Agent C's information comes from a different, independent source.

Respond with ONLY a JSON object in exactly this format, nothing else:
{"answer": "<one of the possible answers, verbatim>",
 "p": {"<answer text>": <integer 0-100>, ...}}
"p" must give a probability for every possible answer listed above, using the
answer text as the key, and the values must sum to 100.

--- response, format required, values illustrative ------------------------
{"answer": "25%", "p": {"10%": 5, "25%": 40, "30%": 5, "15%": 50}}

--- what the scorer derives -----------------------------------------------
world truth       : 15%
decision          : 25%          -> incorrect
p(world truth)    : 50
p(majority claim) : 40
margin, top two   : 10 points -- 50 on 15%, 40 on 25%
coherence         : FAILS -- the highest probability sits on 15%,
                    but the answer selects 25%
```

With four options, coherence is whether the chosen answer carries the highest stated probability. No fixed threshold can stand in for it: chance is 25%, and a judge can put 40% on one option while another holds 50%, so a probability above one half is neither necessary nor sufficient. A record of this shape would be the instrument working as intended: the forced choice returns the majority claim while the judge's own numbers place the world truth highest. That is the strong form of the state H6 names, in which the stated probabilities favour an answer other than the one given. The milder form, a near-level distribution behind a confident-looking answer, passes the coherence flag and is detected instead by the margin between the top two probabilities. A decision-only protocol observes neither. The same case appears in X1 with the note removed and in X5 with a quality statement added, differing only in that block of text.

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